## Appendix B. Dynamo programs for BIM information extraction and model updating

B.1. Extraction of PWP wall information (Section 4.1)

The program developed for extracting PWP information is shown in Fig. 19 and comprises four main steps. First, the Select Model Element node is used to select a PWP wall element that requires optimization design (Fig. 19(a)). Subsequently, the Element.GetLocation node determines the position of the wall baseline (Fig. 19(b)), and the Element.GetParameterByName node obtains the wall height. Finally, the Data.ExportToExcel node saves the extracted PWP wall information in an Excel file (Fig. 19(c)); the output is the wall information described in Section 4.1. Even if the user's workflow is based on other software platforms, the method can be applied by conveniently establishing the PWP walls in the Revit interface.

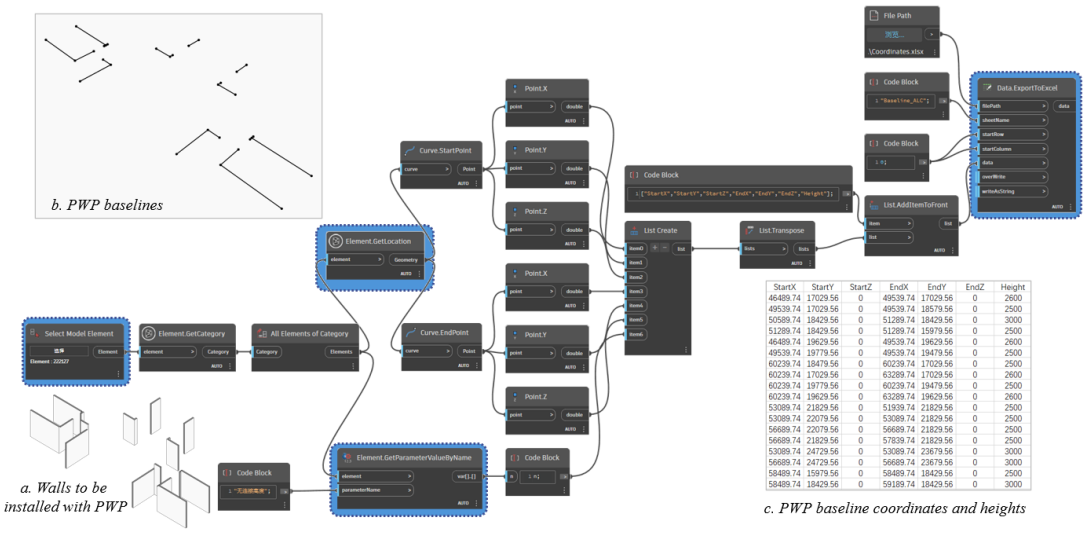



Fig. 19. Dynamo program developed for PWP wall information extraction

B.2. Layout design modeling and coding (Section 4.4.1) and manufacturing combination modeling and coding (Section 4.4.2)

The program for visualizing the layout design outcomes (Fig. 20) executes four main steps. First, two File Path nodes read the storage paths of the two text files generated by the first stage of the nested EA, which contain the information on PWP segments and PWP panels, and another File Path node accesses the path of the NPY file

that stores the PWP body information produced by the second-stage EA. Subsequently, a Python script developed for data reading and processing is executed (the Data Reading and Processing node in Fig. 20); this programmable node allows specific functions to be created within Dynamo using the Python programming language [48]. The Wall.CurveAndHeight node is then employed to define the PWP segments, outlining the PWP panel layout as shown in Fig. 20(a). Finally, the Element.SetParameterByName node writes the pertinent information of each PWP segment into its instance parameters (Fig. 20(b)), namely the three identifiers described in Section 4.4.1. The manufacturing combination program of Section 4.4.2 reads the same files and follows four similar steps, but separately constructs the PWP standard panel BIM model and the PWP body BIM model (Fig. 11 in the main text), in which the dimensional and ID information described in Section 4.4.2 is recorded.

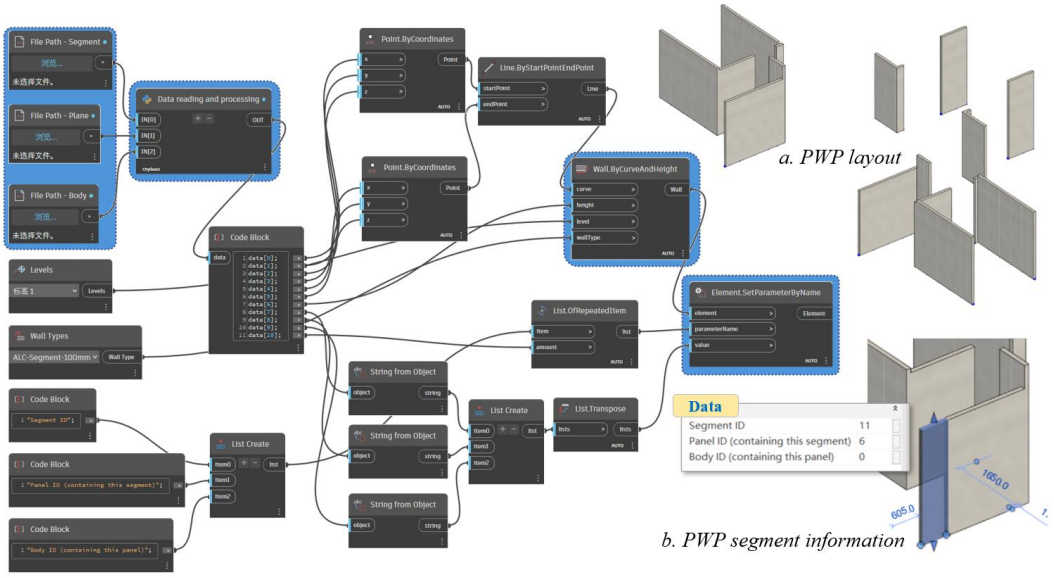



Fig. 20. Dynamo developed program for layout design modeling and coding